

Rational interval designs with polynomial common denominator

Joint consequence of the p-adic and integer-rounding research notes

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Theorem 1. *Fix $A > 0$. If rational nodes $x_1, \dots, x_K$ have a positive common denominator $D \leq r^A$ and give an equal-weight degree- r quadrature for the uniform interval measure, then*

$$\log K \geq r \log r - r \log \log r - O_A(r).$$

Conversely, there is such a quadrature on the grid $i/(r+1)^2$ with

$$\log K \leq r \log r + O(r).$$

Thus, for common denominators bounded by any fixed polynomial of sufficient degree, the optimal number of nodes is $\exp((1 + o(1))r \log r)$. Node repetitions are allowed and are counted in K .

Proof. For every prime $p \nmid D$, all nodes belong to $\mathbb{Z}_p$. The integral-node case of the general-prime theorem in `../rational_designs/p-adic_lower_bound.tex` gives

$$p^{\lfloor r/(p-1) \rfloor} \mid K.$$

Also $\lfloor r/(p-1) \rfloor \geq v_p(r!)$, by Legendre's formula. Therefore

$$\log K \geq \log(r!) - r \sum_{p \mid D} \frac{\log p}{p-1}.$$

Split the last sum at $y = \log r$. The standard elementary Mertens estimate $\sum_{p \leq y} (\log p)/(p-1) = \log y + O(1)$ controls its first part. For the second part use

$$\sum_{\substack{p \mid D \\ p > y}} \frac{\log p}{p-1} \leq \frac{\log D}{y-1} = O_A(1).$$

Stirling's formula proves the lower bound. The real interval restriction was not used for this direction.

For the upper bound, take $N = (r+1)^2$ and positive starting weights $w_i > 1/(8N)$ from the Wilson–Hahn projection. Set

$$L = r! \operatorname{lcm}(1, \dots, r+1), \quad T = 8N \left(2^{r+1} \binom{N+1}{r+1} + 1 \right), \quad K = L \left\lceil \frac{T}{L} \right\rceil.$$

All binomial target moments multiplied by K are integers, and the integer quadrature rounding lemma in `factorial_squared_construction.tex` gives nonnegative integer weights of total K . Repeat grid node i/N according to its integer weight. The resulting equal-weight rule has the asserted properties. Finally $K < T + L$, the standard estimate $\log \operatorname{lcm}(1, \dots, r+1) = O(r)$ gives $\log L = r \log r + O(r)$, and the elementary binomial bound gives $\log T \leq r \log r + O(r)$. $\square$

Proposition 2 (General denominator tradeoff). *Every rational equal-weight degree- r interval rule with K nodes and positive integer common denominator D satisfies*

$$\log K \geq r \log \left(\frac{r}{\max(2, \log D)} \right) - O(r),$$

where the implicit constant is absolute. Consequently, for each fixed $A > 0$, a rule with $K \leq \exp(Ar)$ must satisfy $D \geq \exp(c_A r)$ for some constant $c_A > 0$.

Proof. Use the same estimate $\log K \geq \log(r!) - r \sum_{p|D} (\log p)/(p-1)$, now splitting the prime sum at $y = \max(2, \log D)$. Mertens' estimate bounds the small-prime part by $\log y + O(1)$, while the rest is at most $\log D/(y-1) \leq 2$. Stirling's formula proves the first assertion. If $\log K \leq Ar$, it implies $\log(r/y) \leq A + O(1)$, and hence $y \geq c_A r$. After decreasing c_A if necessary to cover bounded r , this forces $\log D \geq c_A r$ whenever such a rule exists. For $r = 1$ one may have $D = 1$ by using endpoints, so this last assertion is understood for $r \geq 2$ (or asymptotically as $r \rightarrow \infty$). $\square$

Interpretation. The unrestricted rational-design lower bound is exponential, while polynomial common denominators force a factorial-scale number of node copies. The $\exp(O(r))$ rational designs established in `exponential_rational_designs.tex` necessarily have exponentially large common denominator. This statement concerns quadrature nodes, not face-count lower bounds for an arbitrary fair-dice collection.